Matrix Factorization and Regularized Updates in Collaborative Filtering

1 Introduction

Matrix factorization is a crucial technique in collaborative filtering for recommendation systems. The goal is to approximate a user-item interaction matrix $R \in \mathbb{R}^{m\times n}$ using the product of two latent factor matrices $P \in \mathbb{R}^{m\times k}$ and $Q \in \mathbb{R}^{n\times k}$:

$$R \approx PQ^T$$

where k is the number of latent factors.

2 Objective Function

The objective function to minimize is given by:

$$L(P, Q) = \frac{1}{2} \sum_{(i,j)\in\Omega} (R_{ij} - P_i Q_j^T)^2 + \frac{\lambda}{2} \left(\|P\|_F^2 + \|Q\|_F^2\right)$$

where Ω is the set of observed entries in R, and λ is the regularization parameter.

3 Stochastic Gradient Descent (SGD)

In SGD, we update the parameters for each observed entry $(i, j) \in \Omega$ one at a time.

3.1 Gradient of the Loss Function

The gradient of the loss function with respect to P_i and Q_j is computed as follows:

$$\frac{\partial L}{\partial P_i} = -(R_{ij} - P_i Q_j^T)Q_j + \lambda P_i$$

$$\frac{\partial L}{\partial Q_j} = -(R_{ij} - P_i Q_j^T)P_i + \lambda Q_j$$

3.2 Update Steps

For each observed entry, we update P_i and Q_j:

$$P_i \leftarrow P_i - \alpha \left(\frac{\partial L}{\partial P_i} \right)$$

$$Q_j \leftarrow Q_j - \alpha \left(\frac{\partial L}{\partial Q_j} \right)$$

where α is the learning rate.

3.3 Example of SGD

Consider a user-item matrix R:

$$R = \begin{bmatrix} 5 & 3 & 0 \\ 4 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 5 & 4 \end{bmatrix}$$

Assume initial latent factors:

$$P = \begin{bmatrix} 0.2 & 0.8 \\ 0.4 & 0.6 \\ 0.5 & 0.5 \\ 0.7 & 0.3 \end{bmatrix}, \quad Q = \begin{bmatrix} 0.1 & 0.4 & 0.6 \\ 0.9 & 0.2 & 0.5 \end{bmatrix}$$

3.3.1 First Update Step

For the observed entry $(1, 1)$: 1. Calculate the prediction:

Prediction $= P_1 Q_1^T = (0.2, 0.8) \cdot (0.1, 0.9)^T = 0.2 \times 0.1 + 0.8 \times 0.9 = 0.2 + 0.72 = 0.92$

2. Calculate the error:

$$\text{Error} = R_{11} - \text{Prediction} = 5 - 0.92 = 4.08$$

3. Update P_1 and Q_1 (assume $\alpha = 0.01$):

$$\frac{\partial L}{\partial P_1} = -(4.08) \cdot (0.1, 0.9) + 0.01 \cdot P_1$$

$$P_1 \leftarrow P_1 - 0.01 \cdot \frac{\partial L}{\partial P_1}$$

3.3.2 Second Update Step

Now, for the observed entry $(1, 2)$: 1. Calculate the prediction:

Prediction $= P_1 Q_2^T = (0.2, 0.8) \cdot (0.4, 0.2)^T = 0.2 \times 0.4 + 0.8 \times 0.2 = 0.08 + 0.16 = 0.24$

2. Calculate the error:

$$\text{Error} = R_{12} - \text{Prediction} = 3 - 0.24 = 2.76$$

3. Update P_1 and Q_2:

$$\frac{\partial L}{\partial P_1} = -(2.76) \cdot (0.4, 0.2) + 0.01 \cdot P_1$$

$$P_1 \leftarrow P_1 - 0.01 \cdot \frac{\partial L}{\partial P_1}$$

4 Alternating Least Squares (ALS)

In ALS, we fix one matrix and optimize the other iteratively.

4.1 Update for P

Fix Q and solve:

$$P = RQ(Q^T Q + \lambda I)^{-1}$$

4.2 Update for Q

Fix P and solve:

$$Q = R^T P(P^T P + \lambda I)^{-1}$$

5 Example of ALS

Using the same matrix R: 1. Update P:

$$P = RQ(Q^T Q + \lambda I)^{-1}$$

2. Update Q:

$$Q = R^T P(P^T P + \lambda I)^{-1}$$

6 Conclusion

Matrix factorization in collaborative filtering can be solved through either Stochastic Gradient Descent or Alternating Least Squares, both aiming to minimize reconstruction error while applying regularization to avoid overfitting.